CSci5521: Machine Learning Fundamentals

- Final Review

Final Exam

- The exam will be posted on Canvas at 12pm May 9th (Monday), and it is due at 12pm May 11 (Wednesday).
- The exam is open book. However, you mush finish the exam independently without external resources except the course materials (lectures, textbook, etc). You can also email us with any questions before due time.
- Submission procedure is the same as that of the quizzes and the midterm.
- There will be no make-up exams, except in extraordinary and documented circumstances.
- Multiple choice questions + other questions (including extra credit)

Materials

- Homework assignments, quiz questions, practice questions + sample final
- Lecture slides / videos / whiteboard notes
- Textbook

The rest of the slides lists subchapters that are required in exams, suggested key words in each topic, and some exercise questions from the EA book.

Chapter 11. Multilayer Perceptrons

- 11.1 Introduction
- 11.2 The perceptron
- 11.3 Training a perceptron
- 11.4 Learning Boolean functions
- 11.5 Multilayer perceptrons
- 11.7 Backpropagation algorithms
- 11.8 Training procedures

Chapter 11. Multilayer Perceptrons

Key words: Perceptron, multilayer perceptron, stochastic gradient descent, weight update, backpropagation, chain rule, regression, classification

Chapter 13. Kernel Machines

- 13.1 Introduction
- 13.2 Optimal Separating Hyperplane
- 13.3 The Nonseparable Case: Soft Margin Hyperplane
- 13.5 Kernel Trick
- 13.6 Vectorial Kernels
- 13.8 Multiple Kernel Learning
- 13.9 Multiclass Kernel Machines

Chapter 13. Kernel Machines

Key words: support vector machine (SVM), support vector, classification margin, primal/dual problem, hard/soft margin, hinge loss, nonlinear mapping, kernel

Chapter 9. Decision Trees

- 9.1 Introduction
- 9.2 Univariate Trees
- 9.2.1 Classification Trees
- 9.3 Pruning
- Random forest (slides)

Chapter 9. Decision Trees

- Key words: decision tree, attribute, decision tree learning, impurity, entropy, information gain, best split, gini index, misclassification error, overfitting, random forest

Chapter 14. Graphical Models

- 14.1 Introduction
- 14.2 Canonical Cases for Conditional Independence

- Key words: Bayesian network, factorization, conditional independence, causal inference, diagnostic inference

Exercises (optional)

- 11.1, 11.2, 11.4, 11.5
- 13.1, 13.8
- 14.2, 14.3, 14.5

Solutions will be posted on Canvas.